Nonlinear Model Reduction of Intrinsic Beam with QB-POD

RP: Model reduction for very flexible aircraft flight simulator

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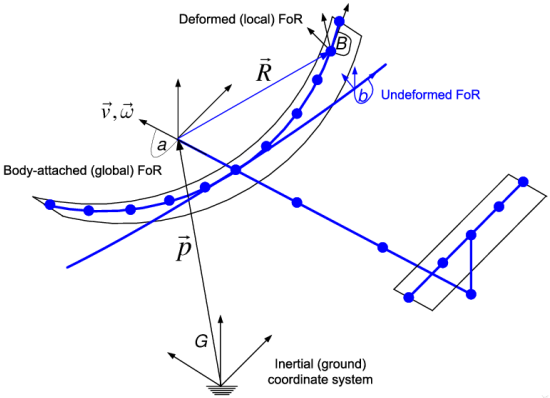

19th March 2026

Motivation

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- HAPS
- Flexible wings
- Research gap

2 Introduction

3 Finite-element model

4 QB-POD

5 Conclusion

Motivation

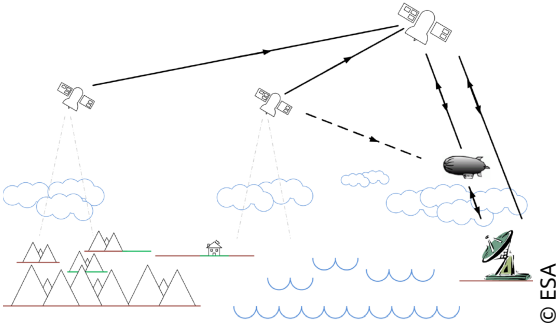
High-altitude pseudo-satellites

HAPS aircraft have applications in

- Surveillance and reconnaissance
- Telecommunications
- Meteorological and environmental monitoring

To sustain flight with solar cells:

- large surface - to capture power.
- light - to be viable off solar power.



Flexible wings

A solution to satisfy both constraints
Flexible wings $\Longrightarrow$ Large wing surface + light.

Consequence on modelling[2]:

- nonlinear structural mechanics,
- unsteady aerodynamics,
- and variable rigid-body mechanics.

Models become **LARGE** and **SLOW**
$\Longrightarrow$ Model order reduction.

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Research gap

- Goizueta[3]: ROM at low speeds using Krylov-subspace model reduction, combining linearised aerodynamic model and linearised beam model.
- Artola[1]: MPC with moving horizon estimation, solves an optimization problem at each time step to compute optimal future control inputs.
- Both linearized the intrinsic beam models.

Linearized beam models inevitable lose fidelity.

Our goal: Reduce the beam model without linearization.

Introduction

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Geometrically exact intrinsic beam

The intrinsinc beam equation were derived by Hodges[4] can be written compactly as[6]

$$\mathcal{M}\dot{\mathbf{x}}_1 - \mathbf{x}_2' - \mathrm{E}\mathbf{x}_2 + \mathcal{L}_1(\mathbf{x}_1)\mathcal{M}\mathbf{x}_1 + \mathcal{L}_2(\mathbf{x}_2)\mathcal{C}\mathbf{x}_2 = \mathbf{f}_{\mathrm{ext}}, \tag{1}$$

$$\mathcal{C}\dot{\mathbf{x}}_2 - \mathbf{x}_1' + \mathrm{E}^{\mathrm{T}}\mathbf{x}_1 - \mathcal{L}_1^{\mathrm{T}}(\mathbf{x}_1)\mathcal{C}\mathbf{x}_2 = \mathbf{0}, \tag{2}$$

The nonlinear term can be separated

$$\mathbf{f}_{\mathrm{NL}}(\mathbf{x}) = \begin{bmatrix} \mathcal{L}_1(x_1)\mathcal{M} & \mathcal{L}_2(x_2)\mathcal{C} \\ & -\mathcal{L}_1^{\mathrm{T}}(x_1)\mathcal{C} \end{bmatrix} \mathbf{x}, \tag{3}$$

and is quadratic as $\mathcal{L}_i$ is linear.

Analytical Jacobian

Using the Jacobian in the numerical solver helps with stability and speed.

The Jacobian of the linear term is trivial

For the nonlinear term, the Jacobian can be found as

$$\nabla \mathbf{f}_{\mathrm{NL}}(\mathbf{x}) = \mathbf{f}_{\mathrm{NL}}(\mathbf{x}) - \begin{bmatrix} \mathcal{L}_2(\mathcal{M}x_1) & \mathcal{L}_1(Cx_2) \\ -\mathcal{L}_1^{\mathsf{T}}(Cx_2) & \end{bmatrix} \tag{4}$$

Proper Orthogonal Decomposition

POD - Modal decomposition and with SVD, truncating at same rank. r, gives the POD basis, Φ.

Data-based - POD basis is computed from samples of FOM
$\implies$ Quality of data is vital.

Optimal basis - for a given reduced order r, POD has lowest $\mathcal{L}_2$-error[7, p. 6].

$$\|\mathbf{A} - \Phi^T \mathbf{A} \Phi\|_{\mathcal{L}_2} \leq \|\mathbf{A} - \mathbf{D}\|_{\mathcal{L}_2} \quad \forall \ \mathrm{rk}(\mathbf{D}) \leq r. \tag{5}$$

POD and Non-linear systems

A linear function expressed as a matrix-vector product can be reduced with POD as

$$f(\mathbf{x}) = \mathbf{A}\mathbf{x} \implies f_r(\mathbf{x}_r) = \mathbf{A}_r\mathbf{x}_r, \quad \mathbf{x}_r = \Phi_{n\times r}^T\mathbf{x}, \ \mathbf{A}_r = \Phi_{n\times r}^T\mathbf{A}\Phi_{n\times r}, \tag{6}$$

so the reduced function scales with $r \ll n$. As for a nonlinear function, applying POD gives

$$f_r(\mathbf{x}_r) = \Phi_{n\times r}^T f(\mathbf{x}) \approx \Phi_{n\times r}^T f(\Phi_{n\times r}\mathbf{x}_r) \tag{7}$$

The nonlinear term is calculated at full-rank, this degrades the performance of POD as $f(\Phi_{n\times r}\tilde{\mathbf{x}}, t)$ still scales with n.

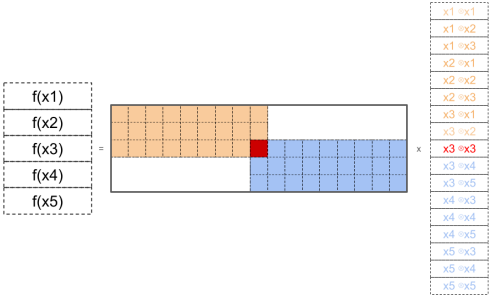
Kronecker formulation

Quadratic forms can be expressed as matrix-vector products using the Kronecker product, called the quadratic-matrix form.

$$\forall f : \mathbb{R}^n \longrightarrow \mathbb{R}^n, \text{ quadratic function}, \exists \mathbf{H} \in \mathbb{R}^{n \times n^2} \tag{8}$$

$$f(\mathbf{x}) = \mathbf{H}(\mathbf{x} \otimes \mathbf{x}) \tag{9}$$

The quadratic nonlinearity in the intrinsic beam equations, Eq. 3, can be expressed this way.



QB-POD

With the nonlinear term in the quadratic-matrix form, it can be reduced to the POD basis directly[5, p. 9]:

$$\mathbf{f}_{\mathrm{NL}}(\mathbf{x}) = \mathbf{H}(\mathbf{x} \otimes \mathbf{x}) = \mathbf{H}(\Phi \mathbf{x}_r \otimes \Phi \mathbf{x}_r) \tag{10}$$

$$\implies \mathbf{f}_{\mathrm{NL},r}(\mathbf{x}_r) = \Phi^t \mathbf{H}(\Phi \otimes \Phi)(\mathbf{x}_r \otimes \mathbf{x}_r) \tag{11}$$

$$= \mathbf{H}_r(\mathbf{x}_r \otimes \mathbf{x}_r), \quad \mathbf{H}_r \equiv \Phi^T \mathbf{H}(\Phi \otimes \Phi) \in \mathbb{R}^{r \times r^2}, \tag{12}$$

Finite-element model

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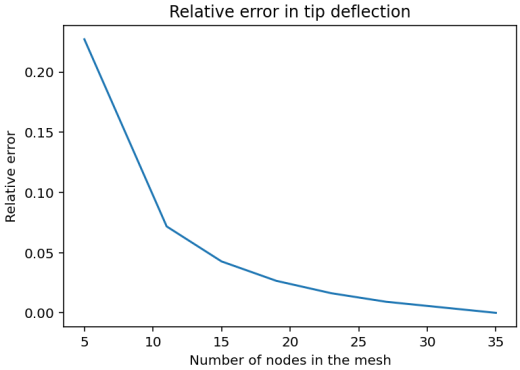
Finite-element beam model

POD is an intrusive reduction technique - needs access to system matrices.
$\implies$ A FEM solver was coded from scratch[a]

- Static and dynamics solver (both formulations)
- Post-processor
- Validation (mesh convergence, linear beam comparison)

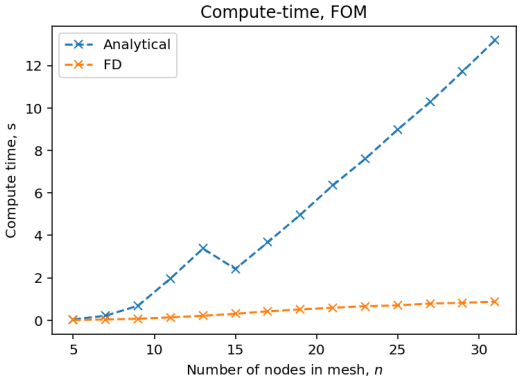


[a] https://github.com/abhyupandey21212/RP, see README.md

Jacobian

Implementing the Jacobian gave a large speed-up.

Bigger mesh $\implies$ bigger speed-up



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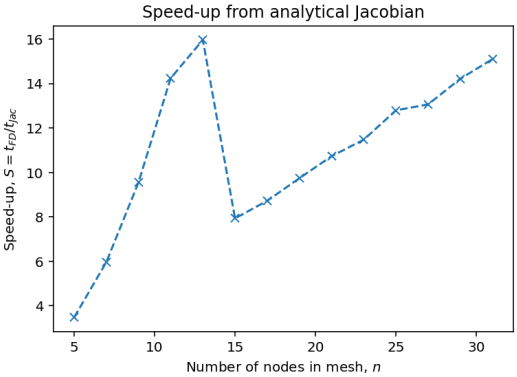

QB-POD

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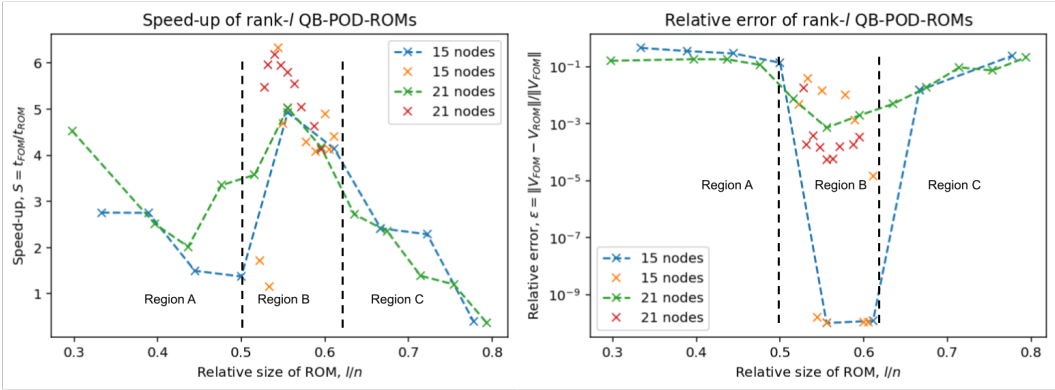
QB-POD


Speed-up of rank-l QB-POD-ROMs
Speed-up, $S = t_{FOM}/t_{ROM}$
Relative size of ROM, l/n
Region A
Region B
Region C
15 nodes
15 nodes
21 nodes
21 nodes
Relative error of rank-l QB-POD-ROMs
Relative error, $\varepsilon = \|V_{FOM} - V_{ROM}\|/\|V_{FOM}\|$
Relative size of ROM, l/n
Region A
Region B
Region C
15 nodes
15 nodes
21 nodes
21 nodes

QB-POD

Region A

- "Too small" - Initially high speed-up, but nonlinear dynamic not captured $\implies$ high error and numerically unstable

Region B

- $\frac{l}{n} \nearrow \implies$ captures nonlinearity, and Jacobian better
- "Goldilocks zone" - Minimum error and maximum speed-up coincide

Region C

- "Too big" - speed-up drops as ROM gets too big, times out, does not converge $\implies$ high error

Speedup

Conclusion

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To conclude

Novel contributions

- The analytical Jacobian. Speed-up of $S > 10$
- QB-POD and Kronecker formulation. Speed-up of $S \approx 6$
- Best QB-POD-ROM at $\frac{l}{n} \approx 0.55$

Further work

- Implementing QB-POD on dynamic solver
- DEIM with QB-POD
- Coupling the aerodynamic model, reducing both

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[1] Marc Artola. *Minimal nonlinear modal aeroelastic descriptions for highly flexible aircraft control*. PhD thesis, Imperial College London, 2021.

[2] Carlos E. S. Cesnik, Cristina Riso, and Divya Sanghi. UNIVERSITY OF MICHIGAN'S NONLINEAR AEROELASTIC SIMULATION TOOLBOX (UM/NAST) WORKSHOP. Workshop presentation, July 2021. Active Aeroelasticity and Structures Research Lab, University of Michigan.

[3] Norberto Goizueta Alfaro. Parametric reduced-order aeroelastic modelling for analysis, dynamic system interpolation and control of flexible aircraft, Apr 2022. URL `http://hdl.handle.net/10044/1/100137`.

[4] Dewey H. Hodges. Geometrically exact, intrinsic theory for dynamics of curved and twisted anisotropic beams. *AIAA Journal*, 47(5):1308–1309, 2009. doi: 10.2514/1.40556. URL `https://doi.org/10.2514/1.40556`.

[5] Boris Kramer and Karen E. Willcox. Nonlinear model order reduction via lifting transformations and proper orthogonal decomposition. *AIAA Journal*, 57(6):2297–2307, 2019. doi: 10.2514/1.J057791. URL `https://doi.org/10.2514/1.J057791`.

[6] Rafael Palacios. Nonlinear normal modes in an intrinsic theory of anisotropic beams. *Journal of Sound and Vibration*, 330(8):1772–1792, 2011. ISSN 0022-460X. doi: https://doi.org/10.1016/j.jsv.2010.10.023. URL https://www.sciencedirect.com/science/article/pii/S0022460X10007078.

[7] Stefan Volkwein. Model reduction using proper orthogonal decomposition. 01 2008.

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